

Zhengkao Zhao

Phone: +1 (312) 340-1603

Email: ichbill27@gmail.com

Website: ichbill.github.io

LinkIn: <https://www.linkedin.com/in/zzhao27/>

RESEARCH INTERESTS

Efficient machine learning, Dataset Distillation, Machine Learning for Science, etc.

EDUCATION

College of Science, Illinois Institute of Technology

Advisor: Yan Yan

Doctor of Philosophy in Computer Science

August, 2020 - present

College of Science, Illinois Institute of Technology

Master of Science in Computer Science

August, 2018 - May, 2020

RESEARCH EXPERIENCE

[1] **Zhengkao Zhao**, Yuzhang Shang, Junyi Wu, Yan Yan, "Dataset Quantization with Active Learning based Adaptive Sampling." In Proceedings of the European Conference on Computer Vision (ECCV), 2024.

[2] **Zhengkao Zhao***, Haoxuan Wang*, Yuzhang Shang, Kai Wang, Yan Yan. "Distilling Long-tailed Datasets." arXiv preprint arXiv:2408.14506, 2024.

[3] **Zhengkao Zhao**, Hao Tang, Joy Wan, Yan Yan. "Monocular Expressive 3D Human Reconstruction of Multiple People." In Proceedings of the ACM International Conference on Multimedia Retrieval (ICMR), 2024.

[4] **Zhengkao Zhao**, Hao Tang, Yan Yan. "Audio-Visual Navigation with Anti-Backtracking." In Proceedings of the International Conference on Pattern Recognition (ICPR), 2024.

[5] **Zhengkao Zhao***, Ye Zhu*, Xiaoguang Zhu, Yuzhang Shang, Yan Yan. "Supplementing Missing Visions via Dialog for Scene Graph Generations." In Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2024.

[6] **Zhengkao Zhao**, Hao Ding, Dawen Cai, Yan Yan. "Gated Multi-scale Attention Transformer for Few-shot Medical Image Segmentation." In Proceedings of the IEEE International Symposium on Biomedical Imaging (ISBI), 2024.

[7] Rathnayake AC Rathnayake*, **Zhengkao Zhao***, Nathan McLaughlin, Wei Li, Yan Yan, Liaohai L Chen, Qian Xie, Christine D Wu, Mathew T Mathew, Rong R Wang. "Machine Learning Enabled Multiplex Detection of Periodontal Pathogens by Surface-Enhanced Raman Spectroscopy." International Journal of Biological Macromolecules, 2024.

INDUSTRIAL EXPERIENCE

Argonne National Laboratory

May, 2023 - August, 2023

Research Aide - Technical, Supervisor: Huihuo Zheng

- Distributed training frameworks evaluation.
- Research on distributed training frameworks: PyTorch DDP, Horovod, and DeepSpeed.
- Conducted experiments across various models using different frameworks to evaluate efficiency.

PROJECT EXPERIENCE

Mouse Behavior Analysis

August, 2022 - present

- Designed and managed the overall project workflow.
- Collaborated closely with medical departments to ensure project alignment and communication.
- Led the data labeling process for accurate annotation of medical datasets.
- Developed and trained a mouse pose estimation model.
- Built and optimized a multi-modality mouse behavior prediction model.
- Created high-quality visualizations.

Rapid Oral Bacteria Detection with Deep Learning

November, 2021 - December, 2022

- Collaborated with the biochemistry department at IIT to discuss progress, address potential issues, and analyze results.
- Preprocessed raw Raman spectra data.
- Developed a model with high accuracy for predicting the existence of bacteria.
- Created visualizations for result analysis and interpretation.

TEACHING EXPERIENCE

Teaching Assistant, Illinois Institute of Technology

September, 2019 - May, 2020

- Co-taught Machine Learning class.

Teaching Assistant, Illinois Institute of Technology

September, 2020 - December, 2020

- Co-taught Science of Programming class.

Teaching Assistant, Illinois Institute of Technology

September, 2021 - December, 2021

- Co-taught Natural Language Processing class.

Teaching Assistant, Illinois Institute of Technology

January, 2023 - May, 2023

- Co-taught Probabilistic Graphical Models class.

COMPETITION EXPERIENCE

2019 Tech to Protect Challenge

October, 2019 - May, 2020

Developer of Critical Access

- Awarded in final stage.
- Demo: <https://github.com/nandavelugoti/critical-access>.
- Designed features and application structure.
- Implemented model part in the Model-View-Controller pattern.
- Implemented features for users to add or remove contacts and groups.

TECHNICAL PROFICIENCY

Tools, Simulation, and Software Platforms

- PyTorch, PyTorch Lightning, Keras, Tensorflow, Hugging Face, etc.

Programming Languages / Scripts

- Python, HTML, C, C++, SQL, etc.

Languages

- English (proficient), Mandarin (native)

EXTRACURRICULAR ACTIVITIES

Summer Research Camp of WhatEvery1Says (WE1S)

June, 2019 - July, 2019

- Trained LDA models using MALLET and Python to attain models with higher accuracies.
- Accomplished mapping visualization using Python Dash and Heroku.